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Multichannel EEG DeepNet for Depression Screening

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***Abstract*—There has been a marked rise in the number of cases of depression across the board in this dire COVID-19 condition. Never before have the issues of mental health, social isolation, and stress related to unemployment been so widely discussed as they are in the COVID-19 era. It is significantly more important to recognize and address intellectual disabilities at some point during pandemics since they can have severely negative consequences if left untreated.**

It is commonly known that 3.8% of people will experience depression, with 5.0% of adults and 5.7% of people over 60 predicted to be afflicted. Around 280 million individuals worldwide suffer from depression. Depression is a complex intellectual disability characterized by a person's poor mood and persistent feelings of hopelessness or misery. This intellectual ataxia-depression can be diagnosed with an electroencephalogram (EEG).

The work focuses on using an automated EEG analysis for the primary screening of Major Depressive Disorder (MDD). The detection method that has been created relies on the combination of 1D convolution and recurrent neural networks (RNNs). The planned structure has been examined using neuroscan on 21 drug-free, symptomatic depressive patients and 24 normal people at the University of Arizona in the United States. With a false positive rate of 0.07 %, the model achieves an accuracy of 84%. RNN is incorporated into the model, which makes it easier to utilize, particularly when the length of the signal is uncertain. The suggested model has a lower temporal complexity and can be used in conjunction with other techniques for MDD screening.

***Index Terms*—EEG, Depression, Deep Neural Network, COVID-19, Detection**

I. Introduction

Depression has long been a topic of interest for analysts in India. Many studies from India over the past 50–60 years have tackled different facets of this often-occurring problem. Among the numerous viewpoints examined were the examination of illness transmission, section and psycho-social hazard factors, neurobiology, symptomatology, comorbidity, assessment, and the impact of loneliness. Globally, COVID-19 has had a devastating impact on people.

Apart from the obvious medical symptoms in situations of infection, it has seriously harmed mental health in general. Like other nations, India shut down the entire country in an effort to control and limit the virus's spread. India has seen an increase in mental health cases during the epidemic, just like the rest of the world. It has been demonstrated that students and healthcare professionals have higher levels of stress, anxiety, and depression than persons in other occupations. Despite the current situation, mental health practitioners' levels of stress, anxiety, and sadness were found to be within normal ranges, demonstrating their ability to remain normal under pressure.

Suicides have occurred as a result of the increased fear and anxiety brought on by COVID-19. According to [1], Chinese nationals experienced severe psychology distress (anxiety, stress, and depression) during COVID-19. Comparably, in another study including Chinese citizens, it was found that psychological distress, including stress, worry, and sadness, was rather widespread and alarming. People's mental health was undoubtedly negatively impacted by pandemics like SARS.

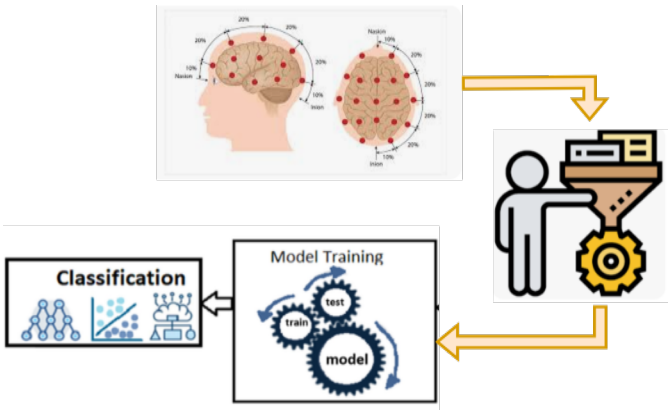


Fig. 1. The general overview of a depression detection system.

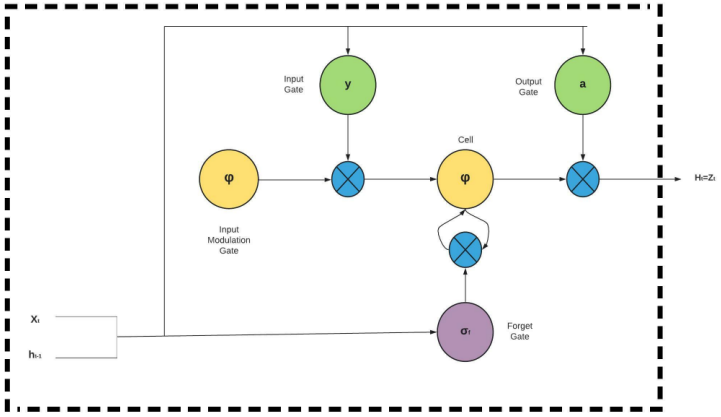


Fig. 2. A general LSTM cell.

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A. Architecture of proposed hybrid model

Depression, as a psychological condition with clinical manifestations such as critical depression and slow reasoning is frequently accompanied by strange mind action and clear emotional turmoil. Therefore, as a method tracking the brain functions, EEG can detect these abnormal activities when compared to non-depressed subjects, patients with depression have hemispheric asymmetry in their brain signals, according to studies. According to a preliminary study carried out in [2], a decrease in the complexity of brain functioning in patients with depression indicates a lower level of interaction with the environment.

The potential of EEG signal as an effective bio marker and diagnostic tool for identifying different neurological disorders has immense scope as the method is non-invasive and economical nature. The brain activity is captured in form of electrical signals, as these are difficult to interpret on a manual base [3]. In recent years, the advance techniques like data mining and Deep Neural Network (DNN) [4], RNN and Convolution Neural Network (CNN)[5].In this work, the author has focused on exploring the recent machine learning techniques to mitigate the issues and effectively analyses the depression conditions.

The complete structure of the article is summarized as: Section.2 provides a brief overview of the existing state of art methods for screening depression by employing EEG signal along with the research gaps and the authors contributions. In Section.3, proposed method is discussed along with its computational complexity. Section.4 discuss the performance of the designed model and comparison of the work with the existing state of art. The work is concluded with future prospects in Section.5.

II. Prior research work and contribution

A. Literature Survey

There have been several attempts for Depression Detection from last decade as listed in Table.II-A. The methods proposed in [6] necessitates the presence of doctors, and the outcomes were subjective. The labeling of data was carried out by collection of linear and nonlinear characteristics collected from artifact-free EEG epochs and then analysed statistically.A correlation based feature selection was adopted in [7], the search method employed was BestFirst. The introduction of computer aid for extraction of EEG properties was employed in [8].There has been works reported on LSTM also [9]. But these work have combined them with the CNN networks. Thus, the overall computational complexity increases.

B. Research gaps addressed in the paper

Through our extensive survey of the various techniques applied for depression detection in literature. The following were the research gaps that we could find:

- There has been methods reported [17] that incorporated usage of handicrafted feature extraction for the detection of disorder. The major drawback of such procedure can be imputation on the performance of designed model.
- The success of CNN model in various fields can not be overlooked. Although the computational complexity and the training time are high. The CNN models deigned in EEG depression detection have presented the advantage of high accuracy [18]. The model consisted of 13 layers.
- The application of RNN to this field is in its rudiment stage. There are a few models designed on LSTM network with promising results [19].
- Deep neural network are a bit complex in nature. Although, the models requires careful training and testing.

C. Authors Contribution

In this article, the author has designed a complete depression detection model as shown in Fig.1.

- In this paper, a type of RNN (i.e LSTM) based Deep Learning model is proposed for depression detection. The model consist of three stacked layer of LSTM with inclusion of dropout and 1D convolution layer.
- The data pre- processed,it includes the denosing using a bandpass filter followed by labelling and normalization of data to avoid biasing of model to EEG recording.
- The model consists of a dropout layer, the major advantages of introducing this layer is prevents all neurons in a layer to optimize their weights in a synchronized fashion. Also, the weights are decorrelated and thus the issue of over-fitting can be addressed effectively.
- The model have used LSTM layer, this eliminates the need of fine tuning of hyper parameters as compared to that required in CNN and DNN. The weight update complexity in LSTM is $O(1)$.
- The proposed design consists of three layered layers of LSTM. The sequence of vectors that each LSTM layer generates is used as an input by subsequent LSTM layers. This system of hidden layers, which records information at several scales, may lead to a more complicated representation of our EEG data.

III. Proposed Methodology

The output of the CNN model is not delivered back to the network at this stage; it only shifts data to different hidden layers. For this reason, CNN algorithms are dreadfully poor at learning consecutive data. This paper proposes the stack LSTM model as a solution to this problem. An element of an RNN, LSTM is used for sequential data learning. The most important feature of LSTM is that, before giving the output to the network, it uses its past knowledge to predict the next element in the sequence. The proposed detailed model is illustrated in Fig.3.

A. Implementation of 1D convolution and Dropout Layer

Convolution is a mathematical tool normally explored to extract features. When the operation is carried out in one dimension, it is referred to as Conv 1D layer. A kernel (weighted filter) is designed that slides over the input signal also known as the shift-compute procedure. The operation can be either casual or non-causal. In this implementation, the author has used non-casual.

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Author	Year	Methodology	Accuracy
Xiaowei Zhang.et.al. [6]	2015	BPNN , kNN	94.2, 92.9
Xiaowei Li.et.al. [7]	2015	kNN	99.1
U. Rajendra Acharya.et.al [8]	2015	CAD	94
Xiaowei Li, et.al [10]	2016	BN, SVM, LR, kNN, RF (RF)	96
Shih-Cheng et.al [11]	2017	KEFB-CSP	81.23
Betul Ay et.al [9]	2019	CNN, LSTM	97.66
S.Dhananjay Kumar et.al [12]	2019	CNN, LSTM	-
Shalini Mahato et.al [13]	2020	SVM, ReliefF	79.19
Ayan Seal, Rishabh et.al [14]	2021	CNN	-
Hui Wen Loh et.al [15]	2021	STFT	99.58
Geetanjali Sharma et.al [16]	2021	CNN	99.1

TABLE I
VARIOUS METHODS PROPOSED IN LITERATURE FOR DETECTION OF DEPRESSION

The input (x) to convolution layer is of length n, the length of kernel(h) is k. The stride s is taken after each convolution operation.

$$y(n) = \begin{cases} \sum_{i=0}^{k} x(n+i)h(i) & \text{if } n = 0 \\ \sum_{i=0}^{k} x(n+i+(s-1))h(i) & \text{otherwise} \end{cases} \quad (1)$$

The output length (O)is not equal to the input layer and is given by

$$O = [\frac{(n-k)}{s}] + 1 \quad (2)$$

The deep learning architectures are now deeper and wider. However, the major drawback is the issue of overfitting and co-adaption. In order to address these issues, the author has introduced the dropout layer.

Co-adaptation poses a serious challenge to large-scale network learning. If all the weights are learned at the same time, it is typical for some links in such a network to have more predictive ability than others. With the dropout layer added, the network's loss function is now defined as follows:

$$L_D = \frac{1}{2}(T - \sum_{i=0}^{k}(\delta_i W_i I_i)^2 \quad (3)$$

the δ is the dropout designed based on Bernoulli's probability distribution function with probability p.

B. Learning model with Stacked LSTM

Long Short-Term Memory Networks, or "LSTMs," are a form of RNN that can comprehend long-term dependencies. Long-term dependency is a problem that LSTMs were designed specifically to solve. Figure2 illustrates the LSTM cell's internal structure, which is consistent with the findings presented in [20]. $y^{<t>}$ denotes the input of the LSTM cell, and $a^{<t>}$ denotes the output. The previous cell output is denoted by $a^{<t-1>}$, while the current and previous cell states are represented by $c^{<t>}$ and $c^{<t-1>}$, respectively. The variables $_f$, , and $_0$, respectively, represent the values of the forget gate, output gate, and update gate. Tanh represents the activation function with element-wise addition , while represents the Hadamard function.

The LSTM model has three main gates: the update gate, the forget gate, and the output gate. The function of each gate is explained below.

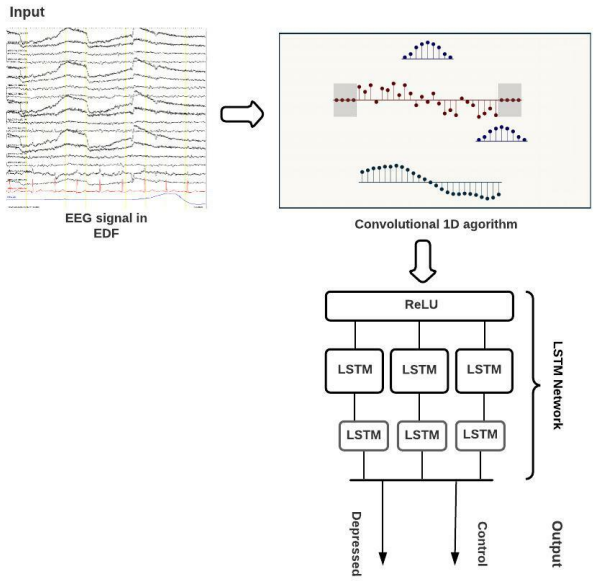


Fig. 3. The complete designed model for primarily screening of depression based on EEG.

- Update gate ($_$)- The key function of the update gate is to decide when to update the current cell state.
- Forget gate ($_f$)- The forget gate fulfils the discard function of the current cell
- Output gate ($_0$)- The control is provided to the output through the output gate.

In this model, the author has used the concept of Stacked LSTM. It is the extension of the existing model where there are multiple hidden LSTM layer where each layer consists of multiple memory cells. Stacking enriches the model performance and prediction ability.

The architecture of proposed model consists of 3 LSTM layer. The details of all the layers incorporated in the deigned model is tabulated in Table.III-B. A convolution operation is formed on the input data vector followed by an activation function i.e. ReLU layer in our case. The next is three consecutive layers are stacked LSTM structure. The last three layers taken are dense and the output neurons for the classification layer. The model uses adam

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Layer (type)	Output Shape	Param #
Conv1D_1 (Conv1D)	(None, 14, 64)	512
dropout (Dropout)	(None, 14, 64)	0
Conv1D_2 (Conv1D)	(None, 12, 32)	6176
Conv1D_3 (Conv1D)	(None, 11, 16)	1040
lstm (LSTM)	(None, 11, 32)	6272
lstm_1 (LSTM)	(None, 11, 32)	8320
lstm_2 (LSTM)	(None, 32)	8320
flatten (Flatten)	(None, 32)	0
Dense_1 (Dense)	(None, 32)	1056
Dense_2 (Dense)	(None, 32) 1056	99.58%
Dense_3 (Dense)	(None, 8) 264	99.1%
Dense_4 (Dense)	(None, 1)	9
Total params: 33,025 Trainable params: 33,025 Non-trainable params: 0		

TABLE II
THE LAYER STRUCTURE OF THE DESIGNED MODEL.

optimizer which is a stochastic gradient descent method with the learning rate of 0.001.

IV. Depression Screening : Results and Analysis

A. Dataset

The EEG signals utilized in this investigation were analyzed at the psychology department of the University of Arizona in the United States.

Through written informed permission, the University of Arizona approved the acquisition of participant data[21]. The individuals were chosen based on their results from the Beck Depression Inventory (BDI) mass survey. The EEG signals of N = 21 sad individuals (14 females) with a BDI score greater than 13 and N = 24 control subjects (14 females) with a BDI score of less than 7 were recorded. The age range of the participants was 18 to 25, and none of them had a history of trauma or seizures, nor did they presently take psychoactive medicine. At a frequency of 500 Hz, 64 scalp electrodes were utilized to record the EEG data.

Fast Fourier Transform was used to extract time-frequency data from the raw EEG signals, and Independent Components Analysis was used to remove artifacts such as eyeblinks. The windowing strategy was used as part of pre-processing since it is a sophisticated way to manage time-series data used in a deep learning-based model. For an 8-second frame, the power spectral density of the healthy and disordered subjects is shown in Figures 6 and 7, respectively.

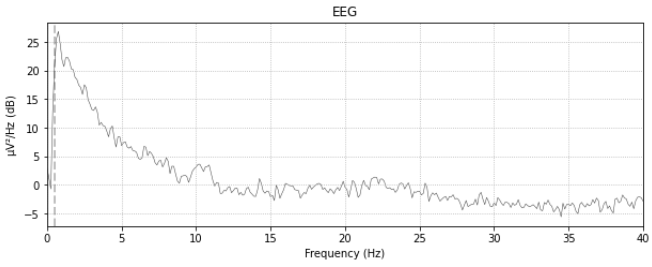


Fig. 4. PSD of a healthy person with effective window size of 8 sec.

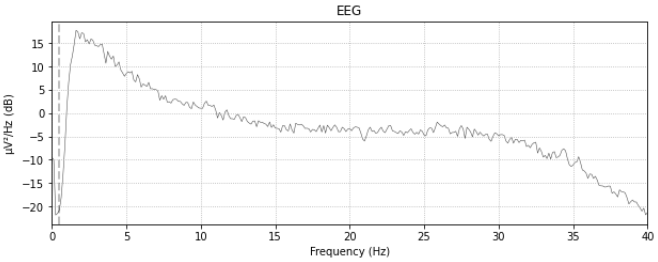


Fig. 5. PSD of a person having major depression disorder with an effective window size of 8 sec.

B. Simulation Platform:

The entire screening model construction and comparison process is completed in the Colaboratory, or "Colab" as it is commonly called. The host's system configuration consists of an Intel Core(TM) i7 processor running Windows 8 (64-bit) with an integrated 2.40 GHz CPU and 8 GB of RAM.

C. Performance analysis:

With a 70% training ratio, the data was divided into training, validation, and testing segments. The remaining 30% was then divided into 10% testing and 20% validation. According to this ratio, there are 2563 and 1000 EEG records for validation and testing, and 9428 records for training. Testing of the model was conducted using a 10% untrained dataset. The dataset has been randomly scrambled to avoid bias. For an epoch of 100, the model obtains an accuracy of 84%, as shown in Fig.6.

However, as Fig.7 illustrates, the model loss is roughly 0.35%. The model took 59 milliseconds to finish an epoch of 100 under the testing conditions described in Section 4.2. Recall and precision percentages are 80% and 85%, respectively, that were noted. With these values, we may investigate the power of stacked LSTM memory cells. The model has taken advantage of two important structural aspects that have fewer complexity when compared to other existing models with a comparable level of accuracy because of the intrinsic capabilities of the anchored LSTM layer.

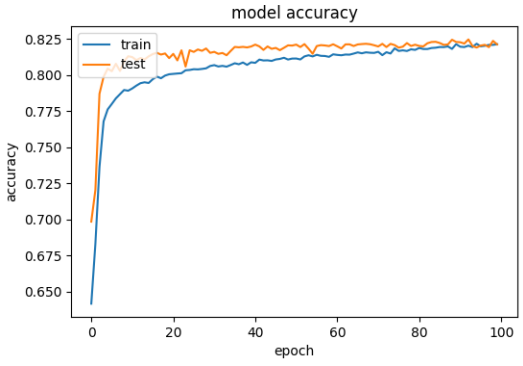


Fig. 6. Accuracy curve of proposed detection model.

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Author & Year	MDD Datatset	Targeted Application	Designed Model	Model Accuracy
Bairy et al,2016 [17]	14 Normal, 16 Depressed	Depression Detection	SVMRBF	88.9
Acharya et al., 2018 [18]	5 Normal, 5 Epileptic	Seizure Detection	CNN (13 Layers)	88.7
Spampinato et al.,2017 [22]	6 Subjects	Discriminate Brain Activity	CNN	86.9
Alhagry et al., 2017 [23]	32 Subjects	Emotion Recognition	CNN-LSTM	85.65
Li et al. ,2019 [24]	27 Normal, 24 Depressed	Depression Detection	ConvNet	85.62
Oh et al.,2019 [25]	14 Normal, 14 schizophrenia	schizophrenia Detection	CNN (11 Layers)	81.26
Proposed Method	24 Normal, 21 Depressed	Depression Detection	**LSTM(3 Layers)**	**84**

TABLE III
COMPARISON OF THE PROPOSED WORK WITH EXISTING STATE OF ART METHODS.

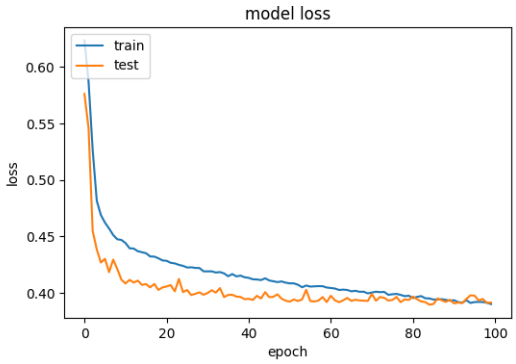


Fig. 7. Loss curve of proposed detection model.

D. Performance Analysis with Existing state of Art Methods:

The comparison of the proposed designed model is done with existing state of art methods as shown in Table.IV-C. The reason for showcasing this compassion's is that with a small dataset the designed networks are too complex. Also, the concept of stacked LSTM structure is still unexplored and has a great potential.

The designed model does not require any additional aid in terms of features or in person to extract the information. The model is self sufficient once's the data has been fed. The main advantage of the designed architecture is that the overall time complexity is less as compared to the other methods for learning sequences.

V. CONCLUSION

The designed model for depression detection introduces the concept of stacked LSTM structure. The stacked not only enhances the feature extraction ability of the model but also improves the prediction ability. The model has been tested on a total of data of 45 patients (24 normal and 21 depressed). The classification accuracy achieved by the model is 84% with a false positive rate of 0.07%. The model is still in its too rudiment stage, the authors plan to improve the accuracy of the model by introducing the ensemble concept.

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